

National University of Computer and Emerging Sciences



Lab Manual 08 Object Oriented Programming (CL1004)

Course Instructor	Mr. Usama Hassan
Lab Instructor (s)	Fareeha Ashfaq Marwa Khan
Section	BDS-2A
Semester	Spring 2022

Department of Computer Science
FAST-NU, Lahore, Pakistan

Objectives

After performing this lab, students shall be able to learn:

- 1: stream insertion and extraction operators overloading (<< , >>)
- 2: Subscript operator overloading []
- 3: association and composition.

TASK 1:

Implement a class called **BiggerInt**. The BiggerInt class will have two data members:

- int* big_int_ ; // Pointer to the int array that holds the big integer
- int int_length_ ; // Variable to store the length of the big integer

While an integer is of 4 bytes in size with a range of -2,147,483,648 to 2,147,483,647. A big integer can store long integer numbers with no size limitation.

You have to implement the following:

Constructors (Default, Overloaded, Copy), Assignment Operator, Destructors, Display Function, Getters and setters.

Moreover, you have to overload the following operators:

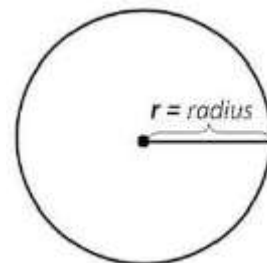
1. []
This operator returns an element in the array only if the index is not out of bounds.
2. **Pre and Post increment**
Increase value of all elements of array by 1.
3. **Pre and Post decrement**
Decrease value of all elements in the array by 1.

TASK 2: (Composition and Association)

Create classes as mentioned below.

```
class Point
{
private:
int* x;
int* y;
public:
//Constructors, Getters and Destructor
};
```

```
class Circle
{
private:
float* radius;
Point* coordinate;
public:
//Constructors, Getters, Destructor
```



```

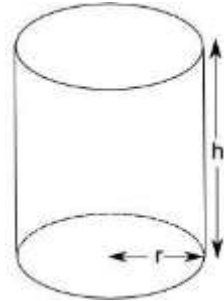
bool CheckOnCircle(Point& p1) //Returns true if point lies in circle,
// false otherwise
};

```

```

class Cylinder
{
private:
int height;
Circle* top; //Circle on top of Cylinder circle* bottom;
//Circle on bottom of Cylinder
public:
//Constructors, Destructors
bool CheckOnCylinder(Point& p1) //Returns true if point lies
//in cylinder, false otherwise
};

```



main() should contains following lines, You can add more code, but these lines should be included:

```

point p1(2,4);
circle c1(5.3,2,3); //Take radius, x, y c1.CheckOnCircle(p1);
cylinder c11(7,5.3,2,9,5.3,2,3); //takes height, parameters for top and bottom circles
c11.CheckOnCylinder(p1);

```

Note

- Follow all the code indentation, naming conventions and code commenting guidelines.